Adaptive Proximal Control for Federated Personalization

Anonymous submission

Abstract

Regularization-based federated personalization methods such as Ditto rely on a fixed proximal coefficient to control how strongly client models remain coupled to a shared global model. This coefficient is not deployment-stable: in our experiments, the best value changes across data regimes and participation rates, so a value tuned before deployment can become stale after conditions shift—a failure we call *coefficient staleness*. We introduce AdaDitto, which treats proximal strength as a bounded online control variable and updates each client's coefficient from a clipped loss-gap signal available during training, while preserving Ditto's objective form and model-update communication pattern. Across CIFAR-100 image classification and Shakespeare next-character prediction, fixed coefficients exhibit distinct regimes; most importantly, on the pathological CIFAR-100 split, the full-participation optimum $\lambda = 0$ becomes stale under 20% participation, where AdaDitto increases coupling online and improves over the frozen baseline by 1.8 points, with larger gains in the client tail. AdaDitto remains below a per-condition oracle that re-tunes the coefficient for each setting, so its contribution is not oracle replacement but reduced dependence on repeated coefficient sweeps under deployment shift.

Code — https://anonymous.4open.science/r/PFLlib-5B8C

Introduction

Federated learning is commonly built around repeated local client optimization and server aggregation, as in FedAvg (McMahan et al. 2017). Under heterogeneous client data, however, local updates can drift away from globally useful directions and aggregation can reflect mismatched local objectives (Karimireddy et al. 2020; Wang et al. 2020). Regularization-based federated learning methods control the relation between local adaptation and global sharing through a proximal term. FedProx (Li et al. 2020) penalizes local drift from the current global model, while Ditto (Li et al. 2021a) trains personalized client models around a shared global model. In both cases, a single scalar coefficient (μ in FedProx, λ in Ditto) governs the strength of this constraint and mediates the trade-off between global consistency and local adaptation.

That scalar is usually selected by an offline sweep. We ask whether a coefficient selected under one deployment condition remains reliable when the participation rate, heterogeneity regime, or active client population changes. Our fixed-Ditto sweeps show that it need not. We call this failure mode *coefficient staleness*: a coefficient that performs well in its tuning condition becomes suboptimal after deployment conditions shift.

This failure appears directly in our fixed-Ditto sweeps. On the CIFAR-100 pathological split, full participation selects pure-local Ditto, with $\lambda = 0$, while reducing participation to 20% moves the best fixed coefficient to nonzero coupling. Freezing the full-participation value therefore produces coefficient staleness under the new participation regime.

We study proximal strength as a bounded online control variable rather than a fixed hyperparameter. From a control perspective, the proximal coefficient acts like a gain: a fixed value trades off stability and responsiveness across operating regimes. But client drift changes during training and across deployment conditions, while the proximal constraint is usually kept static. AdaDitto updates each client's coefficient from a clipped loss-gap signal comparing the received global model's local probe loss to a server reference. The controller uses only scalar quantities available during training, preserves the underlying proximal objective form, and adds one scalar loss report per participating client plus one scalar reference in the server broadcast.

The comparison we target is deployment shift, not hindsight tuning. A fixed Ditto coefficient may be selected under one participation regime and then deployed under another without the luxury of a fresh sweep. AdaDitto is designed for this tune-once setting. In our experiments, it recovers part of the loss caused by a stale full-participation coefficient under CIFAR-100 pathological participation shift, while remaining below fixed Ditto when the latter is re-tuned separately for each condition. The controller should therefore be read as a retuning-reduction mechanism rather than an oracle replacement.

Contributions. We make three contributions. First, we frame proximal coefficient selection as a deployment-stability problem and distinguish per-condition oracle tuning from tune-once deployment. Second, we introduce a bounded loss-gap controller for proximal personalization, instantiated in AdaDitto, that adapts the proximal coefficient without changing the underlying objective form or model-update communication pattern. Third, we show that fixed Ditto coefficients are regime- and participation-dependent across CIFAR-100 and Shakespeare, and that AdaDitto reduces coefficient staleness under CIFAR-100 pathological participation shift while exposing its limits against oracle retuning.

Related Work

Under heterogeneous client data, local federated updates can drift or optimize objectives that are mismatched with the intended global objective. FedAvg (McMahan et al. 2017) provides the canonical local-optimization and server-aggregation baseline. FedProx (Li et al. 2020) adds a proximal penalty to limit local movement from the current global model, SCAFFOLD (Karimireddy et al. 2020) uses control variates to correct client drift, FedNova (Wang et al. 2020) normalizes aggregation to address objective inconsistency, and FedDyn (Acar et al. 2021) introduces dynamic regularization to align local and global solutions. AdaDitto is closest to the regularization line, but adapts the existing Ditto proximal coefficient rather than changing aggregation, adding control variates, or introducing a new global correction term.

Personalized federated learning relaxes the goal of learning one model for all clients. Early federated multi-task methods such as MOCHA (Smith et al. 2017) treat clients as related but distinct tasks. Regularization-based methods such as pFedMe (Dinh, Tran, and Nguyen 2020) and Ditto (Li et al. 2021a) couple personalized models to shared global structure through a penalty term. Other approaches personalize by learning mixtures of local and global models (Deng, Kamani, and Mahdavi 2020), adapting a global initialization (Fallah, Mokhtari, and Ozdaglar 2020), separating shared and client-specific layers (Arivazhagan et al. 2019; Collins et al. 2021), or keeping batch-normalization statistics local under feature shift (Li et al. 2021b). AdaDitto belongs to the regularization-based family: it preserves Ditto's personalized objective and changes only the scalar coupling strength online.

The practical weakness of fixed-λ personalization is that coefficient choice is deployment-dependent. Selecting λ by an offline sweep is costly in federated learning because each candidate may require a distributed run, and noisy partial participation can make validation-based selection unreliable (Kuo et al. 2023; Wang et al. 2023). A fixed coefficient is therefore not only a tuning choice; it is an assumption that the participation regime, heterogeneity structure, and active client population remain close to the condition under which the sweep was performed.

Adaptive federated methods have adjusted server optimizers, aggregation, client learning rates, local training schedules, client participation, and personalization weights in response to observed training signals (Reddi et al. 2021; Kim et al. 2024; Sun et al. 2023; Wu et al. 2023; Shenaj, Belilovsky, and Zanuttigh 2025; Deng, Kamani, and Mahdavi 2020). These methods adapt important parts of the federated pipeline, but they do not directly treat the proximal coefficient as the online control variable. AdaDitto instead adapts the scalar coupling strength between personalized and global models while preserving Ditto's objective form and model-update communication pattern.

Methodology

Interpreting proximal sweeps. The Ditto coefficient λ determines which information source a personalized model trusts during local adaptation. When λ is small, the client model can specialize almost independently of the global model. When λ is large, the client model is pulled toward the shared model and uses global structure as an anchor. Thus a λ sweep is a diagnostic of the coupling regime: $\lambda = 0$ means the best strategy is effectively pure local personalization, while a nonzero optimum means that local adaptation benefits from shared anchoring. A deployment shift is harmful when it changes this coupling regime after λ has already been selected.

We consider clients $\mathcal{C} = \{1, \ldots, N\}$, where client i has empirical objective F_i. At round t, the server broadcasts global parameters $w^{(t)}$ to a sampled subset $\mathcal{S}_t$. Participating clients update the global model for aggregation while maintaining personalized models locally.

Ditto Baseline

Ditto trains each personalized model v_i around the current global model using

$$\min_{v_i} \; F_i(v_i) + \frac{\lambda}{2} \|v_i - w^{(t)}\|^2. \tag{1}$$

The coefficient λ controls the coupling between the personalized and global models: small λ permits stronger local adaptation, while large λ keeps v_i closer to $w^{(t)}$. In standard Ditto, λ is fixed throughout training. AdaDitto replaces it with a client-specific online coefficient $\lambda_i^{(t)}$.

Loss-Gap Controller

At round t, the server broadcasts $w^{(t)}$ and the previous scalar loss reference $L_g^{(t-1)}$. Before local training, each participating client evaluates the received global model on a small local probe minibatch $\mathcal{B}_i^{(t)}$:

$$\hat{L}_i^{(t)} = \frac{1}{|\mathcal{B}_i^{(t)}|} \sum_{(x,y)\in\mathcal{B}_i^{(t)}} \ell(w^{(t)}; x, y). \tag{2}$$

The client compares this probe loss to the server reference using a nonnegative clipped log gap:

$$g_i^{(t)} = \text{clip}\left(\log \frac{\hat{L}_i^{(t)} + \epsilon}{L_g^{(t-1)} + \epsilon}, \; 0, \; \tau \right). \tag{3}$$

The loss gap is used as a stress signal, not as a claim that the current global model is already good for that client. A large positive gap means that the received global model performs worse on client i than on the server reference population. In the participation-shift regimes we target, this gap empirically coincides with conditions where stronger anchoring is

Controller item	Value or rule
Base / range	0.08 with clipping to $[0.05, 3.0]$
Gain / gap cap	$\alpha = 0.8$, $\tau = 0.7$
Smoothing	coefficient EMA $\gamma = 0.3$, reference EMA $\beta = 0.9$
Probe	10% local examples, cap 256, pre-training inference
Communication	one scalar loss up, one scalar reference down

Table 1: Default AdaDitto controller configuration used in all main comparisons.

beneficial. AdaDitto responds by increasing the stabilizing proximal anchor, limiting unconstrained local drift while still allowing personalization through the local objective. The response is deliberately bounded: the controller can increase coupling when the gap rises, but clipping prevents a single noisy probe from dominating training.

The gap is mapped to a target coefficient and smoothed:

$$\lambda_i^{*(t)} = \lambda_{\text{base}} + \alpha g_i^{(t)}, \tag{4}$$

$$\tilde{\lambda}_i^{(t)} = (1-\gamma)\lambda_i^{(t-1)} + \gamma\lambda_i^{*(t)}, \tag{5}$$

$$\lambda_i^{(t)} = \text{clip}(\tilde{\lambda}_i^{(t)}, \lambda_{\min}, \lambda_{\max}). \tag{6}$$

The client then updates its personalized model using Eq. 1 with λ replaced by $\lambda_i^{(t)}$, and returns its usual global-model update together with the scalar probe loss. The server updates the reference for future rounds using a median exponential moving average:

$$L_g^{(t)} = \beta L_g^{(t-1)} + (1-\beta)\cdot \text{median}\big(\{\hat{L}_i^{(t)}\}_{i\in\mathcal{S}_t}\big). \tag{7}$$

AdaDitto changes only the proximal coefficient used for personalized training. The personalized model remains local, the global-model update and aggregation procedure are unchanged, and the extra communication is one scalar probe loss per participating client plus one scalar reference in the server broadcast. Since the gap and final coefficient are clipped, $\lambda_i^{(t)} \in [\lambda_{\min}, \lambda_{\max}]$ for all clients and rounds.

Controller defaults. Unless otherwise stated, all main comparisons use the controller configuration in Table 1, without condition-specific search. We additionally use $\epsilon = 10^{-8}$, a two-round warmup, and at least one probe example.

Experimental Design

We design the experiments around a deployment question: when heterogeneity or participation changes, does a fixed Ditto coefficient remain reliable, and can AdaDitto reduce the need to re-tune it? Each experiment therefore varies one of two stressors: the statistical regime induced by the client partition, and the fraction of clients participating in each round.

The evaluation uses CIFAR-100 (Krizhevsky and Hinton 2009) for image classification and Shakespeare from LEAF (Caldas et al. 2018) for next-character prediction. CIFAR-100 gives controlled regimes: a Dirichlet split with overlapping but unbalanced labels, which we call *practical*, and a label-shard split with stronger client specialization, which we call *pathological*. Shakespeare provides a naturally federated text regime in which clients correspond to speaking roles. The reported metric is personalized-model accuracy. We report personalized-model accuracy because Ditto and AdaDitto optimize client-specific personalized models; tail metrics such as client-unweighted 10th-percentile and worst-client accuracy are used to check whether average gains hide client-level failures.

Table 2 summarizes the evaluation regimes and the role each plays in testing coefficient stability. In the pathological CIFAR-100 split, 20 clients each receive 10 labels, so every class appears on two clients.

Setting	Partition	Participation	Role in evaluation
CIFAR practical	Dirichlet, $\alpha = 0.1$	Full, 50%, 20%	Overlapping-label regime for coefficient stability.
CIFAR pathological	Label shard	Full, 50%, 20%	High-heterogeneity regime for participation-shift stress tests.
Shakespeare	Speaker split	10%; 50% ref.	Text regime checking whether nonzero coupling is beneficial.

Table 2: Evaluation regimes and their roles in the deployment-stability study.

We compare fixed Ditto and AdaDitto under identical models, client sampling, local training budgets, and evaluation code. For fixed Ditto, we distinguish two standards. A *per-condition tuned* baseline re-sweeps λ for each dataset and participation level; this is an oracle-style comparison and is not meant to represent deployment without retuning. A *tune-once* baseline selects λ under full participation and then freezes it under lower participation. This second standard tests coefficient staleness under deployment shift.

The sweep grids and replication roles are listed in Table 3.

Results

The results support a conditional deployment-stability claim. Fixed Ditto coefficients are not stable across regimes and participation rates, and a coefficient tuned under one condition can become stale when participation changes. AdaDitto reduces coefficient staleness under CIFAR-100 pathological participation shift, but it does not replace per-condition oracle retuning. We therefore organize the results by evaluation standard: fixed coefficient sweeps, tune-once deployment, controller behavior, controller sensitivity, loss-gap mechanism, and oracle/null comparisons. We use fixed-λ sweeps to diagnose the preferred coupling regime, per-condition tuned Ditto as a hindsight oracle, tune-once Ditto to measure coefficient staleness, and AdaDitto to test whether online control reduces that staleness without a new sweep.

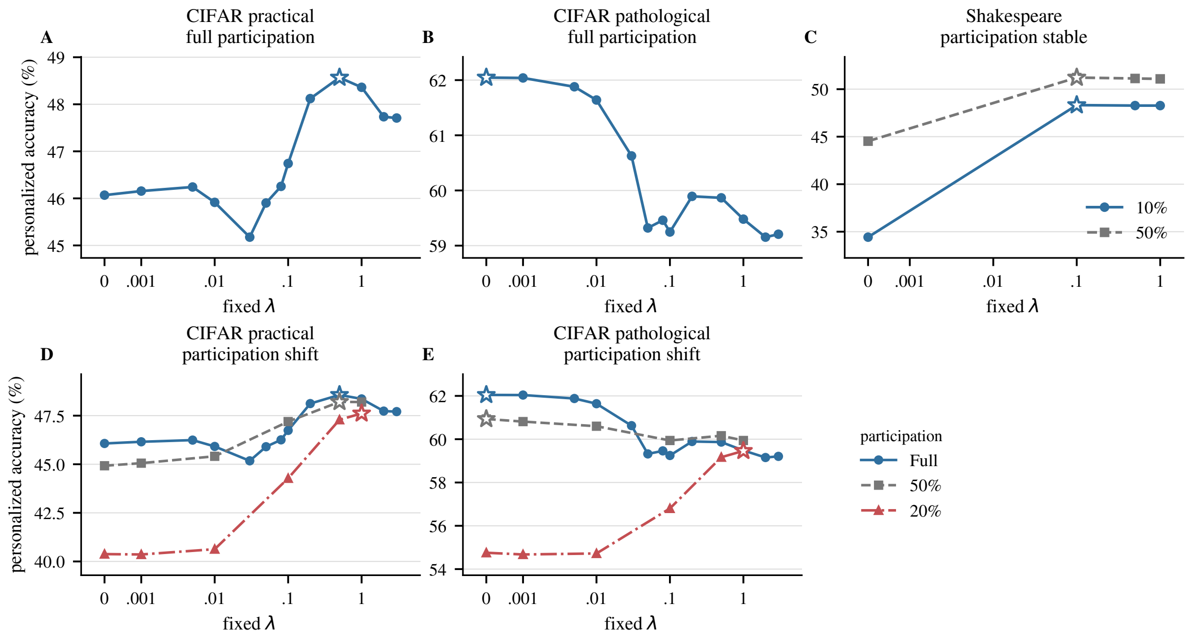

Figure 1: Fixed Ditto coefficients are not deployment-stable. The best λ depends on both heterogeneity regime and participation rate: CIFAR-100 practical favors nonzero coupling, CIFAR-100 pathological under full participation favors $\lambda = 0$, and Shakespeare requires nonzero coupling throughout. The key deployment failure appears on CIFAR-100 pathological, where the best coefficient shifts from $\lambda = 0$ under full participation to nonzero coupling under 20% participation. Stars mark the best fixed λ in each condition.

Experiment role	Grid or comparison rule	Replication
CIFAR full sweeps	$\{0, .001, .005, .01, .03, .05, .08, .1, .2, .5, 1, 2, 3\}$	1 seed
CIFAR partial sweeps	$\lambda \in \{0, 10^{-3}, 10^{-2}, 0.1, 0.5, 1\}$	1 seed
Shakespeare sweeps	$\lambda \in \{0, 0.1, 0.5, 1\}$ at 10% and 50% participation	1 seed
Tune-once shift	tune under full participation, then freeze after shift	3 seeds
CIFAR core comparisons	select λ by per-condition sweep, then evaluate selected Ditto and AdaDitto	5 seeds
Controller stress	vary controller settings around default	3 seeds/config

Table 3: Sweep grids and replication rules. Single-seed sweeps identify coefficient regimes; multi-seed comparisons estimate robustness for the main claims.

Fixed Ditto Coefficients Are Deployment-Dependent

Figure 1 shows that the best fixed Ditto coefficient depends on both the heterogeneity regime and the participation rate. On CIFAR-100 practical, the best fixed coefficient is nonzero. On CIFAR-100 pathological under full participation, the best coefficient is instead $\lambda = 0$, meaning that pure-local personalization is preferred. Shakespeare shows the opposite pattern: $\lambda = 0$ is poor, and nonzero coupling is required.

The key failure mode appears when the participation rate changes. On CIFAR-100 pathological, the full-participation optimum is $\lambda = 0$, but under 20% participation the best fixed coefficient moves to nonzero coupling. Thus a coefficient selected under full participation is not merely suboptimal after the shift; it encodes the wrong coupling regime. This is the coefficient staleness that AdaDitto is designed to address. The other regimes are important negative context. CIFAR-100 practical does not show the same pure-local-to-coupled transition under participation shift, so it should not be used to claim a universal adaptive gain. Shakespeare shows the complementary boundary case: nonzero coupling is consistently useful, but the preferred coupling regime is stable across the tested participation rates. These cases keep the claim narrow: AdaDitto is meant to reduce coefficient staleness when the deployment condition changes the coupling regime, not to guarantee improvement in every heterogeneous federated setting.

AdaDitto Reduces Coefficient Staleness Under Participation Shift

We freeze Ditto's coefficient at the value selected under full participation and evaluate it after reducing participation. On CIFAR-100 pathological, this selects $\lambda = 0$. Under 20% participation, the stale fixed coefficient drops by 7.2 percentage points from its full-participation performance, while AdaDitto drops by 2.8 points and overtakes frozen Ditto by 1.8 points (Figure 2).

Relative to the target condition's best fixed coefficient, the stale $\lambda = 0$ choice incurs 4.7 points of regret, whereas

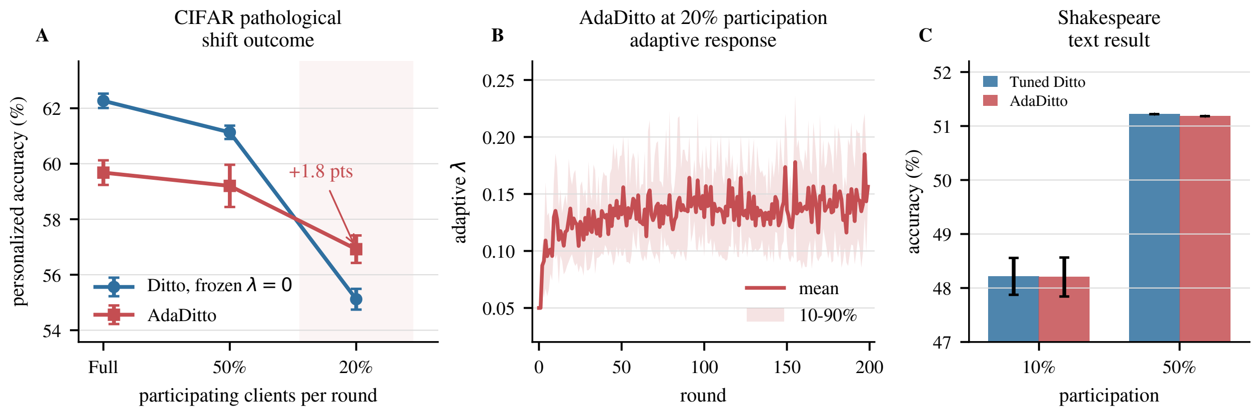

Figure 2: AdaDitto reduces coefficient staleness under participation shift. On CIFAR-100 pathological, the full-participation sweep selects $\lambda = 0$; after shifting to 20% participation, this frozen coefficient becomes stale, while AdaDitto increases coupling online and improves over the tune-once baseline by 1.8 points. The adaptive trajectory shows the learned coefficient rising during training. Shakespeare is a boundary case: AdaDitto matches tuned Ditto because the optimal nonzero coefficient is stable rather than shifted.

Method	Seeds	Mean	p10	Worst
Tune-once Ditto, $\lambda = 0$	3	52.6	42.1	39.9
AdaDitto	3	55.8	49.2	45.3
Ditto oracle, $\lambda = 1$	3	58.7	53.6	50.0

Table 4: Client-tail personalized accuracy (%) on CIFAR-pathological at 20% participation, averaged on the common seed set `0,1,2`. Rows are client-unweighted final accuracies, unlike the headline Figure 2 aggregation.

AdaDitto incurs 2.3 points. The full regret matrix, including cross-regime stress cases, is reported in the supplementary material.

Table 4 shows that the gain extends to the client tail. At 20% participation, AdaDitto improves client-unweighted 10th-percentile accuracy by 7.0 points and worst-client accuracy by 5.3 points over frozen $\lambda = 0$ Ditto.

This tail table is intentionally stricter than the headline aggregate: each client contributes equally within a run, so the statistic asks whether the shift response helps difficult clients rather than only raising the average sampled-client score. The common seed set removes the seed-count mismatch between the tune-once, AdaDitto, and oracle rows, while the oracle row shows that room remains between AdaDitto and a freshly tuned fixed coefficient.

AdaDitto does not improve over the frozen baseline on CIFAR-100 practical, where participation shift does not create the same coupling-regime failure. On Shakespeare, AdaDitto matches tuned Ditto, but the optimal nonzero coefficient is stable, so the result is parity rather than a shift win.

Condition	Best fixed λ	Mean gap	Mean learned λ
CIFAR-path full	0.0	0.013	0.090
CIFAR-path 20%	1.0	0.074	0.135
Shakespeare 10%	> 0	0.003	0.069

Table 5: Mechanism audit connecting the fixed-coefficient sweep, controller signal, and learned coefficient.

The Loss-Gap Signal Tracks Increased Anchoring Need

The adaptive trajectory in Figure 2 is consistent with the fixed-coefficient sweep. On CIFAR-100 pathological under full participation, where the best fixed coefficient is $\lambda = 0$, the post-warmup mean loss gap is 0.013 and the learned coefficient is 0.090. Under 20% participation, where the sweep independently selects stronger coupling, the mean loss gap rises to 0.074 and the learned coefficient rises to 0.135.

Table 5 connects the fixed-coefficient optimum, controller signal, and learned coefficient.

The CIFAR-pathological shift raises both the gap and the learned coefficient; Shakespeare instead shows floor-driven stable coupling.

Controller Tuning Is Not the New Sweep

A possible failure mode is that AdaDitto simply replaces a λ sweep with a brittle controller sweep. Figure 3 tests this at the headline conditions. On CIFAR-100 pathological with 20% participation, all 11 tested controller configurations beat the stale tune-once baseline, and the default configuration is median-ranked rather than cherry-picked. On Shakespeare, all 11 configurations lie in a 0.03-point band and remain within 0.13 points of tuned Ditto.

These scans do not show that controller parameters are irrelevant. They show that the main conclusion does not de-

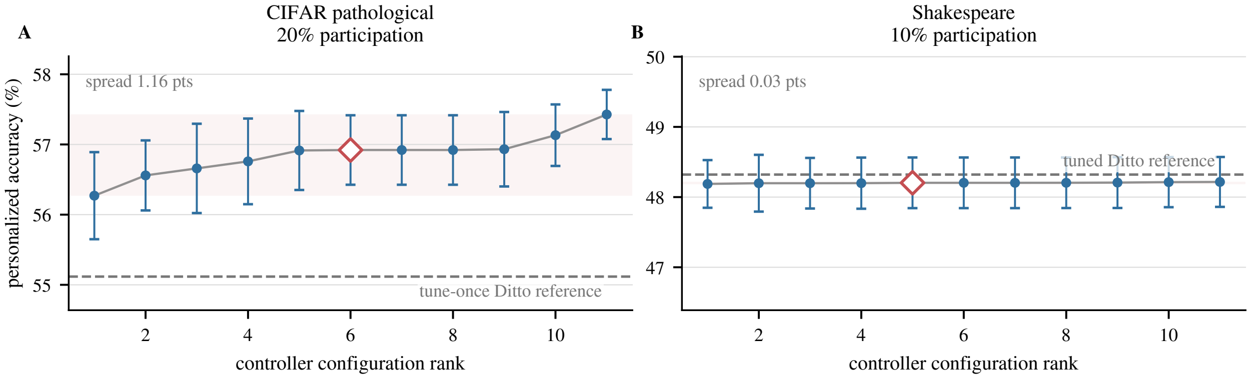

Figure 3: Controller sensitivity at the headline conditions. The white diamond marks the default controller, and the dashed line marks the fixed-Ditto reference for each panel. On CIFAR-100 pathological at 20% participation, all 11 controller configurations beat the stale tune-once baseline, so the shift result does not depend on a cherry-picked default. On Shakespeare, all configurations lie in a 0.03-point band and remain within 0.13 points of tuned Ditto. Both panels average three seeds; error bars show standard deviation.

pend on a lucky default: the CIFAR-pathological shift result survives controller variation, and the Shakespeare parity result remains insensitive.

We also ran controller ablations as a negative control. On CIFAR-pathological, principled variants clustered near 0.597 personalized accuracy, while random adaptive coefficients fell to 0.5863. This separates useful loss-gap adaptation from arbitrary adaptivity under high heterogeneity.

Oracle Retuning, FedProx Nulls, and Shakespeare Parity

The tune-once comparison is deployment-relevant, but it is weaker than a per-condition oracle. When fixed Ditto is re-tuned separately for each participation level, AdaDitto trails the tuned oracle by roughly 1.4–2.6 points on CIFAR-100. AdaDitto should therefore be read as a retuning-reduction mechanism, not as a replacement for exhaustive per-condition tuning when such tuning is available. This calibration matters because the oracle answers a different question from the deployment experiment. The oracle assumes that the target participation regime is already known and that enough distributed runs are available to search it directly. The tune-once setting instead asks what happens when a coefficient chosen before deployment is reused after the operating point changes. AdaDitto's value is in reducing the penalty caused by coefficient staleness, not in beating a hindsight sweep that is allowed to spend a full run on every candidate coefficient.

Per-condition oracle comparisons are reported in Table 6.

FedProx is a negative probe: across tested CIFAR-100 settings, $\mu = 0$ or near-zero remains best. In additional FedProx probes outside the final Ditto protocol, this remains true even with five local epochs and 20% participation, so AdaFedProx has no useful proximal signal to recover. Shakespeare is a boundary case: $\lambda = 0$ is poor, nonzero coupling is stable across 10% and 50% participation, and AdaDitto matches tuned Ditto rather than producing a shift win.

Split	Part.	Ditto	AdaDitto
practical	50%	48.2	46.1
practical	20%	47.0	44.4
pathological	50%	61.0	59.2
pathological	20%	58.7	57.2

Table 6: CIFAR-100 personalized accuracy (%) against per-condition tuned Ditto, averaged over five seeds. The oracle re-tunes the coefficient for each condition; this is the upper bound AdaDitto is not claiming to replace.

The FedProx null results and the Shakespeare boundary case are summarized in Table 7.

Runtime and Communication Overhead

AdaDitto preserves Ditto's model-update communication pattern. Each participating client sends one scalar probe loss, and the server broadcasts one scalar reference loss in the next round; no extra model, gradient, optimizer state, or client-to-client message is introduced. Computation adds only a no-gradient forward pass on a probe minibatch containing 10% of local training examples, capped at 256. Thus the per-round overhead is scalar bookkeeping plus a small local probe, while static retuning requires full distributed runs for each candidate λ.

Reproducibility and Reporting

All reported experiments are implemented as modifications to PFLlib (Zhang et al. 2025). CIFAR-100 is obtained from the public CIFAR distribution; Shakespeare is obtained through the public LEAF preprocessing pipeline (Caldas et al. 2018). Raw datasets are not redistributed. The anonymized code archive is linked in the main paper, and the experiment scripts and run artifacts record the command lines, configuration files, environment snapshots, and per-run

Dataset	Method	Setting	Outcome
CIFAR-100	FedProx	$E = 1$, full participation	small or zero μ best
CIFAR-100	FedProx	$E = 5$, full participation	$\mu = 0$: 63.5; $\mu = 1$: 57.1
CIFAR-100	FedProx	$E = 5$, 20% participation	$\mu = 0$: 23.0; $\mu > 0$: 19.6
CIFAR-100	FedProx	$E = 5$, 20% + systems heterogeneity	$\mu = 0$: 25.9; $\mu = 1$: 22.2
Shakespeare	Ditto	10% / 50%, $\lambda = 0$	34.4 / 44.5
Shakespeare	Ditto	10% / 50%, $\lambda = 0.1$	48.3 / 51.2

Table 7: Null and boundary probes. CIFAR-100 FedProx results show no useful positive-μ regime, while Shakespeare Ditto results show that nonzero coupling is beneficial but stable across 10% and 50% participation. Values are personalized accuracy (%).

Dataset	Model	Rounds	Clients / optimizer
CIFAR practical	CNN	200	20 clients, full/50%/20%, SGD lr 0.01
CIFAR pathological	CNN	200	20 clients, full/50%/20%, SGD lr 0.01
Shakespeare	char-LSTM	40	118 clients, 10%/50%, SGD lr 0.5

Table 8: Final training protocol. All rows use one local epoch, batch size 64, and no weight decay. Ditto and AdaDitto train global and personalized models with the same local learning rate.

metrics used to generate the figures.

The final model and optimization settings are summarized in Table 8.

For multi-seed comparisons, CIFAR core runs use seeds `0,1,2,3,4`. Tune-once shift tests, Shakespeare core comparisons, and controller stress scans use seeds `0,1,2`. Single-seed sweeps are used only to identify coefficient regimes, not as significance claims. Each run applies the seed before data partitioning, model initialization, client sampling, and minibatch shuffling; paired comparisons reuse the same seed set when methods are compared directly.

We do not use null-hypothesis significance tests for the main claims. Instead, we separate roles: single-seed sweeps expose coefficient regimes; multi-seed core comparisons estimate robustness; controller stress scans test whether the default is lucky; and the random-adaptive ablation serves as a negative control. This reporting structure is meant to make the limits of the claim visible: AdaDitto reduces coefficient staleness under deployment shift, but it is not presented as a universal replacement for oracle retuning.

Limitations

The controller's lower bound is load-bearing. When the best fixed coefficient is truly near zero, as in CIFAR-pathological full participation, AdaDitto cannot reduce λ below $\lambda_{\min} = 0.05$ and therefore slightly over-anchors relative to the pure-local optimum. This is a deliberate safety choice, but it means the method should not be read as an exact online optimizer for λ. It is a bounded controller that limits losses caused by coefficient staleness, not a mechanism that recovers every per-condition optimum.

The loss-gap signal is most informative under the CIFAR participation shift, where larger gaps coincide with the fixed-coefficient sweep moving toward stronger anchoring. Shakespeare exposes a different boundary case: the gap is nearly zero, while nonzero coupling remains important. AdaDitto matches tuned Ditto there largely because the lower-bound floor supplies stable coupling, not because the gap detects a shift. This distinction scopes the mechanism: active gap-driven adaptation helps when deployment increases anchoring need, while the floor mainly protects regimes with consistently nonzero coupling.

Finally, the present study uses controlled CIFAR splits and a LEAF Shakespeare check rather than a broad production deployment. We vary participation and heterogeneity, but do not yet study device churn, asynchronous updates, privacy accounting for probe losses, larger language models, or multimodal clients. Those settings may require additional safeguards around probe sampling, coefficient sharing, or privacy-preserving loss reporting.

The main conclusion would weaken under two concrete outcomes. First, if a broader benchmark suite showed that participation shifts rarely change the preferred coupling regime, then coefficient staleness would be less practically important. Second, if privacy or systems constraints made even scalar probe-loss reporting unavailable, then AdaDitto would need a different stress signal. These are deployment questions rather than formatting details: the present evidence supports bounded online proximal control for the tested shifts, not a universal claim about all personalization regimes.

Conclusion

We studied proximal coefficient selection as a deployment-stability problem rather than a one-time tuning problem. Fixed Ditto coefficients can encode the wrong coupling regime after participation shift: on CIFAR-100 pathological, the full-participation choice $\lambda = 0$ becomes stale under 20% participation. AdaDitto uses a bounded loss-gap controller to increase coupling online, reducing the resulting coefficient staleness while preserving the underlying Ditto objective form and model-update communication pattern.

Proximal sweeps diagnose coupling regimes; they do not certify deployment safety.

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